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**SCHOOL OF CIVIL, ENVIRONMENTAL
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PyRestore:

A Configurable Data Processing Framework for Street-View Perception
Analysis

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<https://github.com/GuojialeGeographer/2026GProject.git>

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Abstract

Street-view imagery has become an important data source for urban research. Most large-scale studies use image segmentation to measure physical scene composition, such as greenery, sky and building coverage. These measures do not directly describe subjective qualities such as restorativeness, perceived safety or environmental quality. As vision-language models (VLMs) develop, researchers can use natural-language prompts to turn these visual perceptions into analyzable indicators.

To support urban analysis, researchers must still combine a georeferenced image inventory, physical scene features and VLM outputs into a dataset that can be used in GIS, statistical analysis or later modelling. PyRestore provides a configurable data-processing framework for this step. The researcher supplies a standardized street-view manifest and a YAML task configuration that states the research question, prompt, output fields and optional human reference. The framework then applies the common processing steps and produces one analysis table per image observation, containing spatial coordinates, CV-derived physical indicators and VLM-derived perception indicators.

Restorative-quality scoring is the principal application. For 465 Shenzhen street-view images, the four-dimension mean prediction reached a Pearson correlation of $r = 0.337$ with human ratings, with field-level correlations from 0.108 to 0.572. This task used a general zero-shot prompt without task-specific fine-tuning or retrieval-based prior knowledge. The result is therefore a baseline for this configuration, not a direct substitute for a population-specific survey scale. Researchers can add domain prompts, examples or their own references in the task configuration to adapt the framework to other perception studies, but each configuration requires separate validation.

PyRestore does not train a new perception model. It provides a reusable processing framework that extends street-view analysis beyond physical segmentation and produces spatial datasets that support both physical-environment and visual-perception research.

Keywords: street-view imagery; geoinformatics; computer vision; vision-language model; urban perception; data-processing framework; task configuration; spatial analysis

1. Introduction

Street-view imagery records streets, vegetation, buildings, transport infrastructure and other visible conditions from a pedestrian perspective, which makes it useful for urban analysis [1]. One major line of research measures physical scene composition. Li et al. derived a modified Green View Index from street-view images [2], while Cityscapes and SegFormer support scalable pixel-level extraction of urban composition [3, 4]. These approaches answer what is visible in an image, but they do not directly answer how people perceive the environment.

Urban-perception research addresses this gap. Place Pulse 2.0 collected pairwise judgements on Safe, Lively, Beautiful, Wealthy, Boring and Depressing, enabling large-scale modelling of subjective urban impressions [5]. VLMs now allow researchers to ask perception questions in natural language and to derive indicators such as restorativeness, safety, beauty or renewal need from street-view imagery. This creates opportunities for a wider range of human-centred urban studies.

Researchers still need a practical way to turn raw street-view inventories into analysis-ready datasets. Existing work has shown the potential of VLMs for restorative-quality assessment [6], prompt-defined streetscape mapping [7], neighbourhood assessment [8] and temporal renewal interpretation [9]. However, a researcher usually has to prepare the image inputs, run the model, organize its output fields, join coordinates and export a new dataset for each study. These repeated steps limit the reuse of street-view perception analysis.

PyRestore addresses this engineering gap. It starts with a standardized street-view manifest, in which each row provides an image path and geographic coordinates. The researcher then uses a task configuration file to state the question, prompt, output fields and optional human reference. PyRestore processes the imagery and produces a dataset in which each street-view observation is associated with its coordinates, CV-derived physical scene indicators and VLM-derived perception indicators. The dataset can be used directly for GIS mapping, statistical testing or further modelling.

The framework must accommodate three practical differences across studies. Perception tasks use different prompts and output fields; some tasks use one image, whereas change research requires comparable before-and-after imagery; and model outputs must remain matched to the image and reference data selected by the researcher. PyRestore places study-specific choices in the task configuration and retains image processing, field organization, data joining and geospatial export as shared operations.

Figure 1 summarizes this workflow. A standardized manifest and task configuration identify the images and outputs for an analysis. CV and VLM components extract physical and perception indicators, which the framework then organizes into an analysis-ready dataset. Optional validation and GIS products extend the same dataset. Caching, failure records and provenance support reliable repeated processing, but they are implementation mechanisms rather than the central research goal.

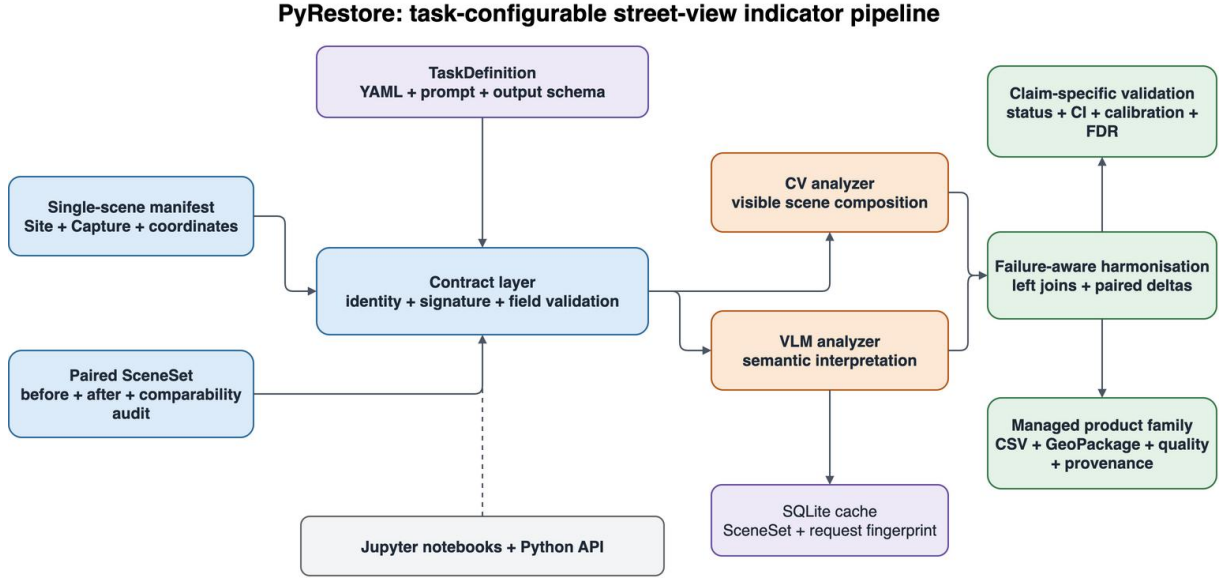


Figure 1. PyRestore starts with a standardized street-view manifest and task configuration, extracts physical-scene and visual-perception indicators, and organizes them into datasets for validation, GIS mapping and further analysis. The same workflow supports both single images and before-and-after image pairs.

The project makes three contributions. First, it implements a configurable street-view data-processing framework, so that researchers can define a perception analysis in a task file rather than rewrite the processing pipeline for each question (Sections 4 and 5). Second, it organizes CV-derived physical indicators and VLM-derived perception indicators into one spatial dataset, with quality checks, validation and GIS export (Sections 4 and 5). Third, it evaluates the framework through a 465-record Shenzhen restorative-quality analysis and tests its adaptation to six-dimension perception and paired-image tasks (Section 6).

2. Related Work and Comparative View

2.1 Street-view imagery and physical scene indicators

Biljecki and Ito reviewed street-view imagery across urban analytics and GIS and documented its use in greenery, transport, health, perception and spatial-data applications [1]. The review also emphasised dependence on commercial imagery services and the resulting access and reproducibility risks. Physical indicators derived from street-view pixels provide one response to the scale of these collections. Li et al. showed how visible greenery could be expressed through a street-level Green View Index [2]. Later semantic approaches classify each pixel into a fixed urban taxonomy. Cityscapes established a labelled urban-scene benchmark [3], and SegFormer combined a hierarchical Transformer encoder with a lightweight decoder for semantic segmentation [4]. PyRestore uses a Cityscapes-fine-tuned SegFormer configuration to estimate class shares and derives GVI, sky share and enclosure from those shares. These values remain model-derived physical-composition estimates rather than field measurements.

2.2 Human-reference perception and restorative quality

Urban-perception research adds a semantic layer that cannot be reduced to pixel shares. Dubey et al. described Place Pulse 2.0, which contains 110,988 images and approximately 1.17 million pairwise comparisons across six perceptual attributes [5]. The six labels used by PyRestore follow that vocabulary, but the Shenzhen reference values used here come from the local Ma and Kwan data package rather than from a new Place Pulse sample. Restorative-quality assessment draws on Attention Restoration Theory [10] and instruments such as PRS-11 [11]. Ma and Kwan combined human PRS references, VLM descriptions and retrieval-based prior knowledge to assess restorative quality in Shenzhen [6]. PyRestore does not reproduce their complete prior-knowledge method. It reuses a matched historical zero-shot score table and retains its original prompt and model information rather than relabelling it as a current run.

2.3 Configurable VLM streetscape workflows

Recent work has made street-scene semantics configurable through natural language. SAGAI combines open street geometry, street-view imagery, a vision-language model and automated mapping; its publisher abstract describes task adaptation through prompt modification and output aggregation to points and street segments [7]. StreetLens similarly presents a researcher-configurable workflow that grounds neighbourhood assessment in domain-informed questions and produces both objective and subjective annotations [8]. Configurable prompting and geospatial outputs are therefore not unique to PyRestore. PyRestore focuses on organizing these capabilities into a reusable data-processing workflow: tasks may change the question, prompt and output fields while still producing consistent, analysis-ready spatial datasets.

2.4 Temporal imagery and renewal interpretation

Temporal street-view data support questions that a single image cannot answer. Bai et al. analysed 105,332 paired Shanghai street-view observations from 2017 and 2022 with an expert-knowledge-guided VLM workflow [9]. Their design motivates PyRestore's paired task: comparable before-after observations, explicit nuisance changes, structured intervention types, evidence text and an uncertain outcome. PyRestore implements and tests the corresponding paired-image processing rules; due to data limitations, it does not reproduce the Shanghai empirical analysis. Its four-pair case contains synthetic solid-colour images and injected analyzer responses, demonstrating that the paired mechanism can run; it provides no evidence of detection accuracy in real renewal settings, and this module will be extended further once suitable data becomes available.

2.5 Spatial inference and interoperable research software

Local Indicators of Spatial Association provide observation-level measures of local spatial clustering [12]. When a local statistic is tested at many locations, however, unadjusted thresholds increase the chance of false positive declarations. PyRestore applies the Benjamini-Hochberg false-discovery-rate procedure to two-sided local permutation pseudo-p values [13]. It also exports GeoPackage, an OGC-defined SQLite-based geospatial exchange format [14]. These choices fit a broader research-software objective: FAIR4RS distinguishes software from static data and emphasises findability, accessibility, interoperability and reusability for executable, versioned research objects [15].

2.6 Comparative scope

Work	Primary task	Configurable semantics	Input signature	Failure/provenance emphasis	Validation and GIS output
Ma and Kwan [6]	Restorative quality	PRS-specific prompt and retrieved prior knowledge	Single scene	Research-run metadata	Human comparison and Shenzhen analysis
SAGAI [7]	Prompt-defined streetscape assessment	Prompt configurable	Single scene	Workflow reproducibility	Point/street mapping and task-level evaluation

Work	Primary task	Configurable semantics	Input signature	Failure/provenance emphasis	Validation and GIS output
StreetLens [8]	Neighbourhood audit	Domain-informed questions	Single scene	Researcher configuration	Structured annotations and human-centred assessment
Bai et al. [9]	Urban-renewal interpretation	Expert taxonomy and constraints	Temporal pair	Study-specific workflow	Human validation and city-scale analysis
PyRestore	Configurable street-view analysis dataset	Question, prompt, fields and reference configurable	Single or audited pair	Shared data processing and provenance	Quality checks, validation, FDR screening and GeoPackage

3. Innovation of the Project

Restorative-quality, six-dimension perception and renewal studies require different questions, output fields and image inputs. Separate scripts for each study would repeatedly implement image loading, model calls, field organization, spatial joining and GIS export. Adding a perception indicator would then require a code change rather than a new analysis configuration. PyRestore centralizes these repeated operations in one framework and lets researchers specify the study-specific content in a task configuration file.

The project's innovation is to turn the process from street-view imagery to an analysis-ready dataset into an efficient, reproducible data-processing framework, rather than a new foundation model or segmentation algorithm. TaskDefinition is the code representation of a task configuration: it defines the prompt, output fields, input form and optional validation. SceneSet represents the image input for an analysis, either one image or a before-and-after pair. The framework reads these settings and applies the same processing workflow, so PRS, Place Pulse and renewal studies do not require separate pipelines in the framework code.

4. Project Development and Used Tools

4.1 Project scope and requirements

4.1.1 Scope

The project includes manifest and pair validation, task loading, CV/VLM analyzers, response caching, harmonisation, continuous and categorical validation, spatial screening, GeoPackage/CSV export, tests and notebooks. It excludes image acquisition, model training, causal evaluation, policy-effect estimation, justice analysis, a web application and a separate command-line interface. Source and licence conditions prevent redistribution of the raw imagery. The public interaction surface consists of Python functions and four executable notebooks.

4.1.2 Functional requirements

The complete requirements catalogue appears in docs/REQUIREMENTS.md. The principal requirements are to validate stable Site and Capture identities, support single and paired SceneSets, re-audit externally supplied pairs, validate task and analyzer schemas, preserve every processing or exclusion outcome, separate CV and VLM evidence, report claim-specific validation states, control local spatial false discovery and export a managed family of interoperable products. Each requirement maps to a module and at least one automated test.

4.1.3 Non-functional requirements

The non-functional requirements cover reproducibility, failure transparency, privacy, portability, testability and usability. The framework records environment, task and software versions, input hashes and request provenance. It preserves status and error fields through manifest-left joins, disables remote image transmission

by default, resolves paths relative to source manifests and supports offline tests through injected analyzers. Notebooks and a user manual provide the public workflow, while validation and spatial states constrain interpretation.

4.2 Development path

The framework and the task file divide responsibility along one line. Identity validation, per-record failure handling, GeoPackage export and notebook interaction stay in the framework, because every task needs them unchanged. The prompt, output-field types, derived values, validation mapping, paired-image rules and spatial field stay in the task file, because they are specific to what is being measured. Runtime configuration is kept separate again, covering provider, model, cache, image preprocessing, CV model and output settings, so that credentials and infrastructure settings never mix with task semantics.

Three task definitions then exercised the design. `prs11.yaml` represents a four-dimension single-scene interpretation with a derived mean. `sixdim.yaml` represents six independent perception-oriented scores and demonstrates a new semantic task without framework edits. `renewal.yaml` changes both output semantics and input signature. It requires an audited temporal pair, returns a categorical decision, constrained intervention types, nuisance differences, confidence and visible evidence, and declares CV delta fields. This sequence follows a rule-of-three development strategy: abstraction followed concrete task pressure rather than preceding it. A fourth definition, `renewal_priority.yaml`, was added later under the same stabilised contract to check generality further rather than to drive the design (Section 6.5).

The hardening phase focused on cases that could silently invalidate results. A paired cache key initially covered only the baseline image, which allowed two pairs sharing a baseline to collide. The final `SceneSet` key hashes every image in order. Precomputed tables initially bypassed live-response validation and could be relabelled with the current model and prompt. The final pipeline validates precomputed rows against the task schema, preserves the actual source model/hash and records whether the source matches the current task/request contract. Spatial screening initially accepted constant scores and unadjusted local pseudo-p values. The final implementation rejects zero variance, constructs KNN weights after projection and applies FDR correction.

4.3 Input and task configuration

4.3.1 Site and Capture identity

The manifest contains one row per capture. Required fields are `site_id`, `capture_id`, `image_path`, `lon` and `lat`. Site identifies a spatial location that may own multiple captures; Capture identifies one image observation. This separation prevents file names from becoming an accidental temporal or spatial model. Optional provenance fields include capture date, heading, camera, source and licence. Missing provenance remains missing rather than being inferred from pixels or file names.

The loader trims identifiers, rejects missing or duplicate capture IDs, converts coordinates to numeric values and enforces longitude/latitude ranges. Relative image paths resolve from the manifest location. A legacy four-column PyRestore table can be adapted by treating its image identifier as both site and capture identity. This compatibility path supports historical evidence without forcing the new contract to adopt the old ambiguity.

4.3.2 Paired-image input

A paired task operates on two captures from the same site. `make_pairs` forms candidates within each site and orders known dates so that the earlier capture becomes the baseline. Each candidate records the before/after IDs and paths, coordinates, dates, distance, heading difference, comparability status and exclusion reason. Candidate IDs are SHA-256-derived from site and capture identities, so unrelated row insertion does not renumber existing pairs.

The pair audit checks known and distinct dates, a task-defined distance tolerance and a task-defined heading tolerance. External pair tables are not trusted solely because they contain `comparability_status = ok`; the pipeline recomputes machine-checkable conditions. Excluded candidates remain in `observations.csv`, while only accepted pairs reach analyzers. This distinction allows a report to explain how many candidates existed and why others did not support a temporal comparison.

4.3.3 Task configuration file (TaskDefinition)

A task file includes a stable ID, version, input signature and prompt. Output fields may be bounded scores, numbers, non-empty strings, constrained categories or lists whose items can also be category-constrained. Derived fields currently support means and sums over declared numeric inputs. Reserved names prevent a semantic field from overwriting operational fields such as `vlm_status`, `vlm_model` or `cache_hit`.

Task loading resolves the prompt path relative to the task file and validates every declaration before execution. Category options must be unique strings. Derived fields must reference existing numeric fields. Paired tasks can declare comparability tolerances and CV delta columns. Validation declarations map predicted columns to reference columns or define positive, negative and abstention values. Spatial declarations select the numerical output, neighbour count, significance threshold and p-value adjustment.

4.4 Analyzer layer

4.4.1 CV physical estimates

The live CV path uses a Cityscapes-fine-tuned SegFormer model [3, 4]. Heavy dependencies load lazily, allowing the package, tests and precomputed workflows to run without PyTorch. The model predicts a class label for every pixel; PyRestore converts class counts to image shares and prefixes class fields with `seg_`. GVI is the vegetation share, SVF is the sky share, and enclosure sums building, wall and fence shares. OpenCV-based colour and Canny edge statistics provide additional visible-image descriptors.

These outputs are physical-composition estimates in a declared image and model coordinate system. They are not field measurements of canopy, three-dimensional sky visibility or building geometry. Panorama projection, camera position, season and segmentation-domain shift affect the values. For this reason, the framework preserves the segmenter identity and keeps CV outputs separate from VLM interpretations.

The Shenzhen cases use precomputed CV values from the local reference package. The pipeline validates their status fields and verifies that normalized composition fields fall between zero and one. This execution mode is recorded as precomputed, not live.

4.4.2 VLM semantic interpretations

The VLM path builds one multimodal user message containing the task prompt followed by every SceneSet image in declared order. Images are converted to RGB, downscaled so their longest side does not exceed the configured maximum and encoded as JPEG data URLs. Remote transmission is blocked unless `allow_remote_images` is explicitly true. The returned text is decoded as JSON and validated field by field. A failed response produces a row with `vlm_status = error` and a diagnostic message rather than terminating unrelated records.

Live VLM calls in this project use gpt-5.4-mini through an OpenAI-compatible endpoint. Model availability is not assumed: an initially requested deployment returned `DeploymentNotFound` for both chat-completion and response requests during earlier operational testing, so every live batch is preceded by a smoke test on a small number of real images (`data/live_smoke/`) before the full run begins. This operational sequence is documented as model compatibility testing, not as a model benchmark.

4.4.3 Cache and request identity

SQLite stores validated VLM response objects. Its primary key combines a SceneSet cache key, task ID, model and request fingerprint. The SceneSet component hashes every image file in order. The request fingerprint covers the task prompt hash, model, provider ID, maximum image size, JPEG quality and completion-token limit. The task prompt hash itself covers task identity/version, input signature, prompt text, temperature, output fields and derived fields.

The cache uses write-ahead logging, a busy timeout and explicit connection closing. A hit avoids a remote call, while an offline miss becomes an error row. Content addressing prevents a changed image from inheriting the result of an unchanged file name. Including both images protects paired tasks in which one baseline participates in more than one candidate pair.

VLM request, cache and failure-preservation states

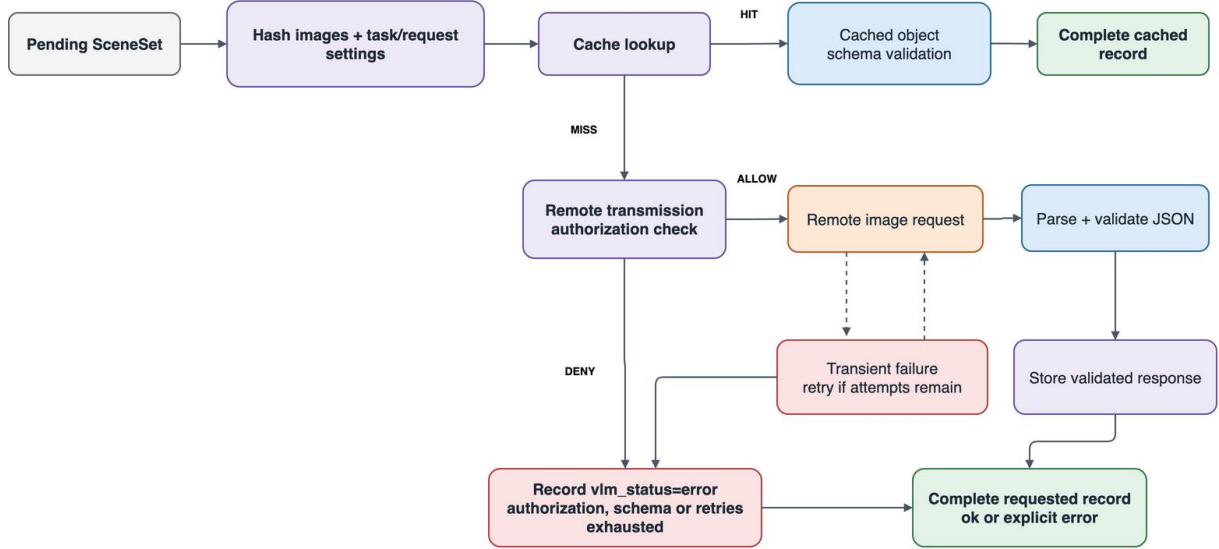


Figure 2. A request completes through a schema-valid cache hit, a validated remote response or an explicit error record. Authorization, parsing and retry failure do not remove the input observation.

4.5 Harmonisation and product publication

The harmonisation layer treats the manifest or accepted pair table as the left-hand contract. CV and VLM tables join by canonical identity with one-to-one validation. Missing analyzer results therefore remain visible as null fields rather than disappearing through an inner join. Human labels receive a `human_` prefix when attached to an analysis table so that they cannot be confused with predicted values.

For paired tasks, capture-level CV results join twice. Fields acquire `_before` and `_after` suffixes, and declared deltas are computed as after minus before. VLM outputs remain pair-level fields. The baseline coordinates define the product point because the comparability contract restricts spatial separation; before and after coordinates remain available for audit.

Before export, PyRestore computes full SHA-256 identities for referenced input images. Shared CSV and GeoPackage products use paths relative to the output directory when possible. The software writes a complete run to a sibling staging directory before publishing managed files. `run_manifest.json` records the successful file set and lets a later successful run remove stale products, such as a map that is no longer valid. A failed computation leaves the previous successful output untouched.

4.6 Validation layer

4.6.1 Continuous agreement

Continuous validation inner-joins predictions and references on a declared ID and reports the joined and valid sample sizes. Each mapping receives a target-level state. Missing columns, zero overlap, fewer than two valid pairs and constant inputs do not become validated rows. Valid mappings report Pearson correlation, its two-sided p-value and 95% confidence interval, together with Spearman rank correlation and its p-value.

Correlation does not test calibration. The validation therefore also reports predicted and human means, standard deviations, ranges, numbers of unique model values, mean absolute error and mean bias. A model can preserve rank ordering while compressing the scale or shifting all values.

4.6.2 Classification and abstention

Paired categorical validation distinguishes positive, negative and abstention values. Missing reference values, missing predictions and values outside the declared categories receive separate counts. Selective precision, recall, F1 and accuracy use evaluable non-abstained rows. Coverage reports the fraction of valid references

with evaluable predictions, while overall recall and accuracy treat unresolved outcomes as incorrect. The current paired case has no independent labels and therefore remains not_validated.

4.7 Spatial analysis and GIS products

PyRestore converts valid longitude/latitude rows to an EPSG:4326 GeoDataFrame. Before KNN construction, geographic coordinates are projected to an estimated local UTM CRS so neighbour relationships are based on planar distances rather than decimal degrees. The implementation rejects missing geometry, non-finite values, too few unique locations and zero-variance indicators.

Global and Local Moran statistics use 999 permutations with a fixed seed. Local permutation pseudo-p values are represented as two-sided values and adjusted through the Benjamini-Hochberg procedure [12, 13]. Only observations that remain below the declared threshold and occupy the Low-Low quadrant receive `candidate_low_cluster = true`. This decision changed the inherited Shenzhen interpretation: weak global clustering remains, but the 40 unadjusted PRS candidates reported by PyRestore do not survive the current FDR rule.

Analysis-ready products use the OGC GeoPackage format [14]. QGIS 3.34 LTR renders the maps in EPSG:32649 with a common fixed zero-to-one score scale. Each map reports the global statistic and the result of FDR-corrected local screening.

4.8 Development tools

Tool	Role in PyRestore
Python 3.10+	Package and notebook runtime
pandas and NumPy	Tabular contracts, joins and numerical preparation
GeoPandas, Shapely and Pyogrio	Geometry creation and GeoPackage I/O
PyTorch, Transformers and Pillow	Optional live segmentation and image preprocessing
OpenAI-compatible Python client	VLM transport abstraction
SQLite	Content-addressed response cache
SciPy	Correlation tests and confidence intervals
Libpysal and esda	KNN weights and Moran statistics
QGIS 3.34 LTR	Cartography and GIS inspection
Jupyter	Primary executable user interface
pytest, coverage and Ruff	Automated behavior, coverage and static quality gates
uv	Dependency lock and reproducible environment synchronisation
Draw.io	Editable specification diagrams

5. Specification Documents

5.1 Component and data-flow specification

The framework overview in Figure 1 is both a component diagram and a high-level data-flow specification. Blue components own input and task contracts; orange components generate analyzer evidence; purple components manage task/request control; green components create validation and product evidence. The arrows represent data or orchestration dependencies, not inheritance. Notebooks and Python functions are clients of the contract layer rather than alternative implementations.

At runtime, a TaskDefinition and an input source converge in the contract layer. The resulting SceneSets feed CV and VLM analyzers. Analyzer outputs converge only during failure-aware harmonisation. Validation

consumes the harmonised table and an optional task-appropriate reference. Managed products include analyzer tables as well as the final indicator table so a downstream user can distinguish raw analyzer outcomes from joined values.

5.2 BPMN execution process

PyRestore BPMN-style execution process

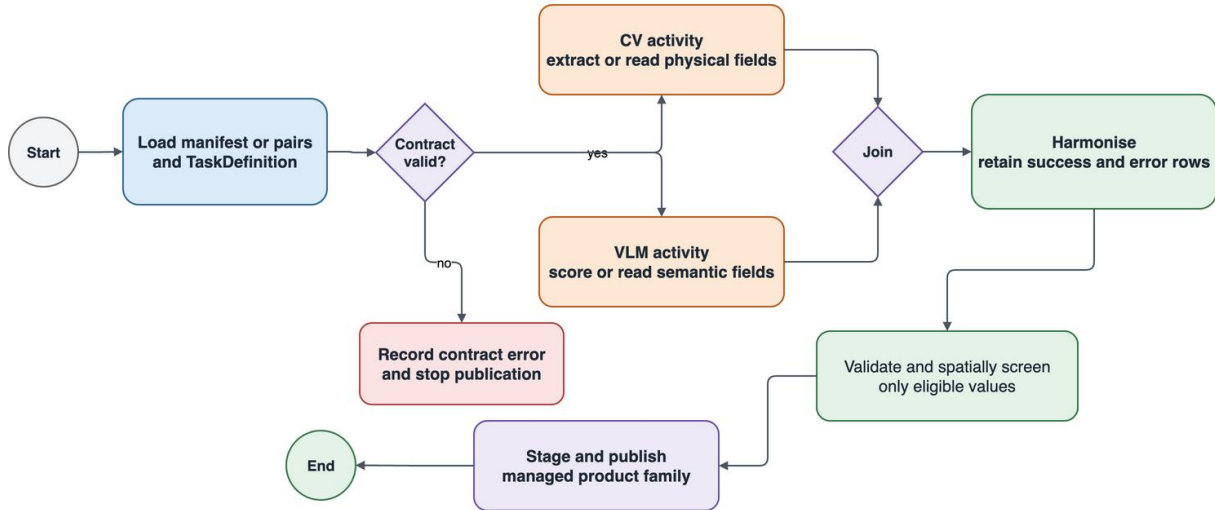


Figure 3. The process starts by loading a manifest or pair table with a TaskDefinition. An invalid contract records an error and prevents publication. A valid contract enters the CV and VLM activities, whose outputs converge during harmonisation. Validation and spatial screening operate only on eligible values before staged publication creates the managed product set.

Contract-level and record-level failures have different effects. An invalid input contract stops publication because the run cannot establish valid identities. An analyzer error affects only its SceneSet and remains visible in the harmonised table. Validation and spatial screening are conditional: a task without a reference or an eligible spatial field still produces indicators, but the corresponding result remains explicitly unvalidated.

5.3 Domain and product data model

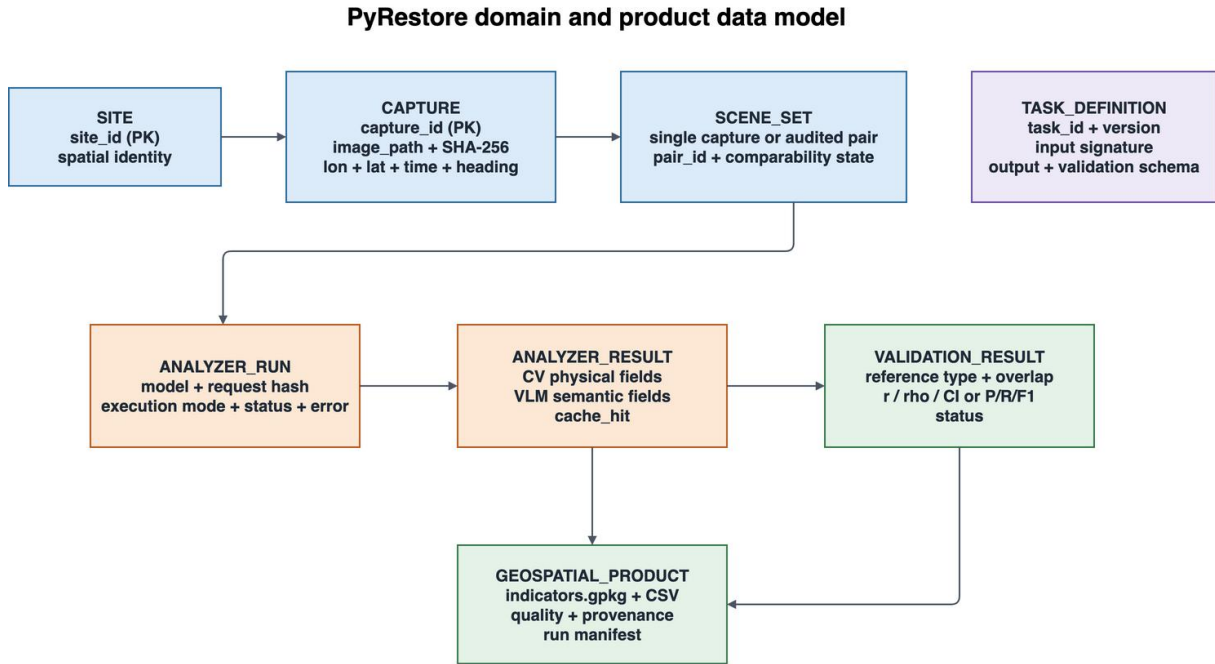


Figure 4. This ER-style domain and product data model shows how stable Site and Capture identities form SceneSets. Task declarations configure analyzer runs whose results feed optional validation and managed geospatial products. TaskDefinition is a declaration object rather than a row in the run database.

Site owns spatial identity. Capture owns image-observation identity and its path, content hash, coordinates and optional acquisition metadata. SceneSet groups evidence according to the task signature. An AnalyzerRun identifies an operational attempt through model, request hash, execution mode and state. AnalyzerResult contains one evidence family and cache information. ValidationResult records a claim-specific comparison rather than mutating the analyzer result. GeospatialProduct packages harmonised values and run-level quality/provenance.

5.4 State-transition specification

Figure 2 documents the VLM request state machine. A pending SceneSet first receives a complete fingerprint. A schema-valid cache object can complete the record without network access. A miss reaches the remote path only after the authorization check. Transient transport failures can retry; schema, authorization or exhausted-retry failures converge on an explicit error state. The error state is terminal for that record but not for the batch.

The state design separates availability from validity. Receiving text from a provider does not mean the task succeeded; only a response that satisfies the field contract can enter the cache as a successful result. Conversely, an error is not an absent observation. Its record remains part of processing coverage and quality reporting.

5.5 UML module and dependency model

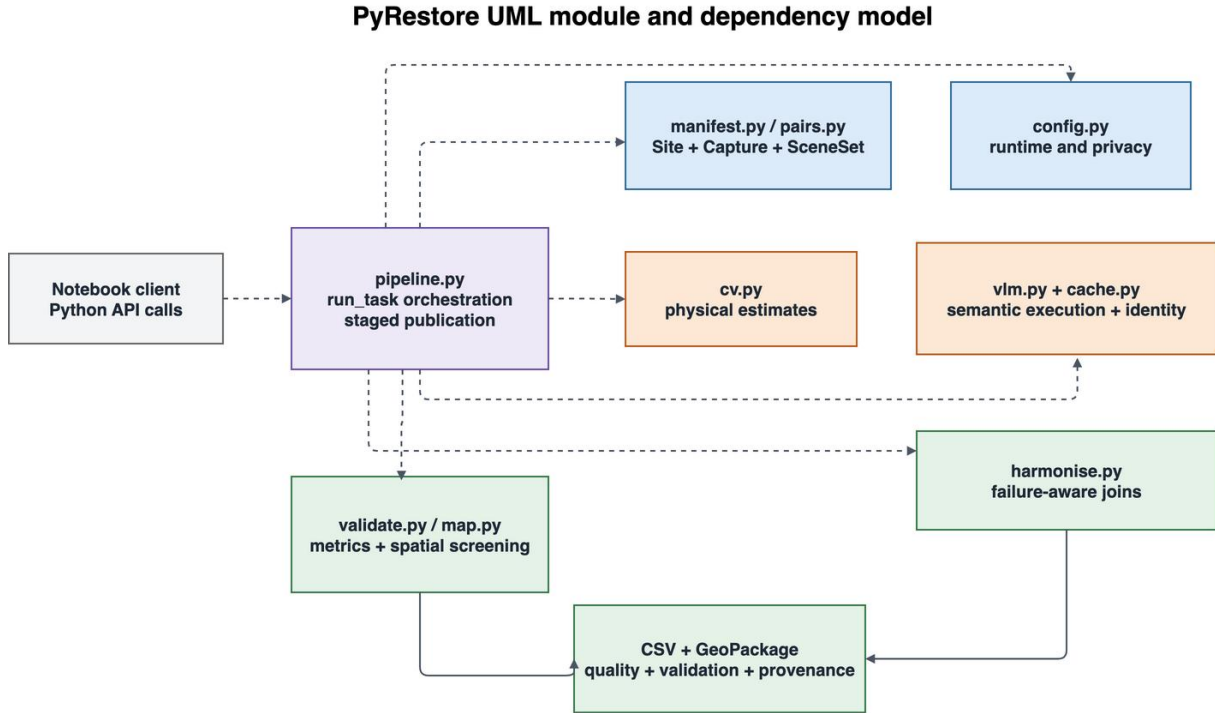


Figure 5. Notebooks call the public API in pipeline.py, which coordinates manifest, pairing and runtime-configuration modules with the CV, VLM and cache components. harmonise.py combines analyzer outputs before validate.py and map.py produce statistical and spatial results. The final outputs include CSV and GeoPackage data together with quality, validation and provenance records.

The module boundaries separate task semantics, runtime configuration and execution logic. manifest.py and pairs.py manage input identity, config.py manages runtime and privacy settings, and the analyzer modules manage physical or semantic results and request identity. pipeline.py owns orchestration and staged publication without embedding task-specific field names.

5.6 Python interaction sequence

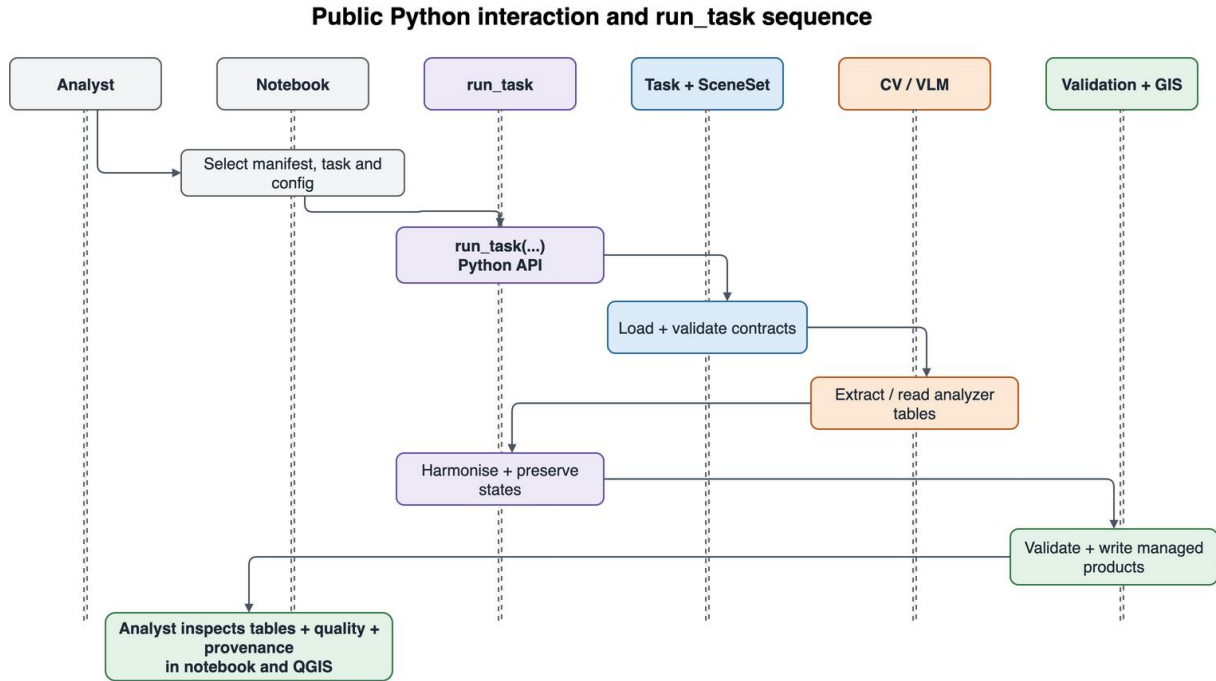


Figure 6. The analyst selects inputs in a notebook and invokes one Python orchestration path. Contract loading, analyzer execution, harmonisation, validation and managed publication return inspectable tables and paths to the same notebook and to QGIS.

The interaction begins with a notebook selecting a manifest, task and runtime configuration. `run_task` validates those contracts before analyzer execution. It then harmonises independent analyzer tables, performs optional validation and writes managed products. The returned dictionary contains in-memory tables, quality, provenance, validation status and final paths. No separate command-line orchestration exists.

5.7 Use cases

The first use case processes an unlabelled single-scene manifest. It produces indicators with `not_validated` status and does not require a ground-truth column. The second adds a human reference and produces target-level continuous validation. The third derives or loads temporal pairs, re-audits comparability and computes before-after context. The fourth applies spatial analysis only to a valid numerical field with sufficient variance and locations. Across all cases, the analyst inspects quality and provenance before interpreting scores.

5.8 Data dictionaries

Manifest and pair fields

Field	Type	Meaning
site_id	text	Stable spatial-site identity
capture_id	text	Unique image-observation identity
image_path	text	Absolute input path internally; relocatable path in shared products
image_sha256	text	Full input-byte identity
lon, lat	real	WGS84 longitude and latitude
captured_at	text/date	Provider-supplied capture time; never inferred
heading	real	Optional viewing direction in degrees

Field	Type	Meaning
pair_id	text	Stable hash-derived pair identity
before_capture_id, after_capture_id	text	Ordered temporal capture identities
distance_m, heading_diff_deg	real	Comparability diagnostics
comparability_status	category	ok, excluded or unaudited before pipeline re-audit
exclusion_reason	text	First failed comparability rule or external exclusion

Analyzer and product fields

Field family	Meaning
seg_*, GVI, SVF, enclosure	CV-derived visible composition estimates
vlm_<task field>	Task-defined semantic interpretations
cv_status, vlm_status	Per-record ok or error state
cv_error, vlm_error	Diagnostic error text
vlm_model, prompt_hash, request_hash	Model and request identities
cache_hit	Whether the VLM response was reused
human_*	Optional human-reference columns attached for analysis
local_l, lisa_q, lisa_p, lisa_p_adj	Local spatial statistics and adjusted pseudo-p values
candidate_low_cluster	FDR-significant Low-Low flag under the declared weights

Run-level products

Product	Purpose
observations.csv	Complete input inventory, including excluded pair candidates
cv_features.csv	Capture-level CV outputs and states
vlm_scores.csv	SceneSet-level VLM outputs, model identity and states
indicators.csv/.gpkg	Harmonised analysis-ready attributes and point geometry
quality_report.csv	Input, coordinate and analyzer coverage counts
validation_report.csv	Reference type, overlap, metrics, confidence intervals and states
provenance.csv	Task, software, source/request, execution and data-status metadata
run_manifest.json	File ownership for the latest successful staged run

5.9 User-experience specification

Jupyter notebooks form the primary interface. Notebook 01 introduces the single-scene contract and product family. Notebook 02 replays the Shenzhen PRS validation. Notebook 03 shows that a new semantic task changes task files rather than framework modules. Notebook 04 demonstrates paired evidence and audit states. The user manual exposes the same Python functions for analysts who want to write their own notebooks.

QGIS is the downstream GIS interface. The user opens indicators.gpkg or spatial_screening.gpkg, checks the CRS and declared field, applies the shared score classes and reads the statistical notes with the map. HTML maps remain optional exploratory products, while final cartography is produced with PyQGIS.

6. Results

6.1 Software acceptance

The locked PyRestore 0.2.1 environment completed 75 offline automated tests. Branch-aware coverage reached 77.09%, above the configured 75% gate, and Ruff reported no static errors in the package, tests, scripts or notebooks. The suite covers manifest and pair contracts, stable pair identity, output-schema parsing, nested categories, request fingerprints, cache isolation, precomputed-table validation, missing-value behavior, classification abstention, continuous-validation states, zero-variance spatial rejection, FDR p-values, staged publication and GeoPackage round trips.

The tests deliberately avoid API credentials and the optional PyTorch stack. Fake clients and extractors exercise orchestration without turning endpoint availability into a software-test dependency. A separate authorized provider run supplied live evidence for the six-dimension VLM path. The optional live SegFormer path remains outside the default CI evidence; the Shenzhen CV fields are precomputed.

All four notebooks executed without hidden cell errors in the locked environment. They demonstrate the workflow and preserve an inspectable execution record; they do not independently reproduce every statistical result in this section.

6.2 Shenzhen matching and data preparation

The source human-reference table contained 566 complete rows for Safe, Lively, Beautiful, Wealthy, Boring and Depressing, each originally expressed on a zero-to-five scale. The local image directory contained 7,307 indexed PNG files. Image and reference file names encoded coordinates at different floating-point precision, so the preparation script rounded longitude and latitude to five decimal places. This rule matched 465 reference rows to image files and left 101 unmatched, for 82.2% availability coverage.

The matched manifest stores reference-table IDs as stable capture IDs and points to local image files without copying them into PyRestore. Human six-dimension values were divided by five to create a zero-to-one validation scale consistent with the task contract. Precomputed segmentation shares supplied the CV channel. The output contains input SHA-256 values, source and licence notes and a machine-readable matching report.

The matched subset is an availability sample. No evidence shows that the 101 unmatched rows are missing at random. The results therefore characterise the 465 matched records and must not be presented as complete coverage of the original reference dataset.

6.3 Restorative-quality case (PRS-11): validation

The restorative-quality case reuses historical VLM scores for the 465 matched Shenzhen records rather than issuing new live requests. This choice preserves the exact source identity attached to those scores. The source model is gpt-5.4-mini and the source prompt hash is 35128c8cfbadece8. PyRestore 0.2.1 validates the historical table against the current product contract while retaining that identity. Because the current task definition uses a different versioned request contract, provenance reports `precomputed_contract_match = false` and does not present the historical scores as a new run.

PRS field	Pearson r	95% CI	Spearman rho	n
Being-away	0.572	[0.507, 0.630]	0.580	465
Coherence	0.108	[0.017, 0.197]	0.125	465
Scope	0.372	[0.291, 0.448]	0.360	465
Fascination	0.162	[0.072, 0.249]	0.179	465
Four-dimension mean	0.337	[0.254, 0.415]	0.341	465

Being-away has the strongest agreement and Coherence the weakest; the four-dimension mean reaches $r = 0.337$. These values reproduce the historical correlation results within the current validation implementation,

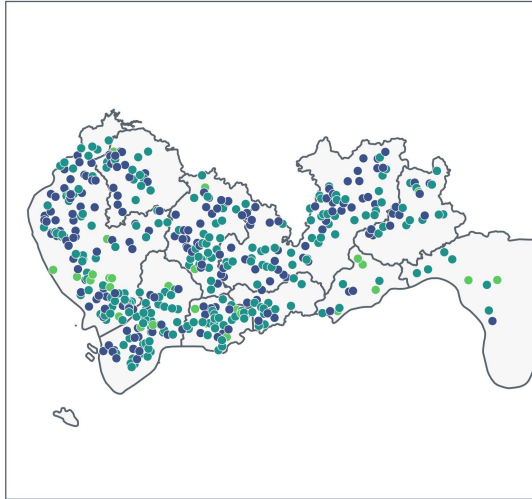
now with confidence intervals, an explicit reference type and a target-level validation state rather than one aggregate number.

6.4 Spatial pattern and QGIS product

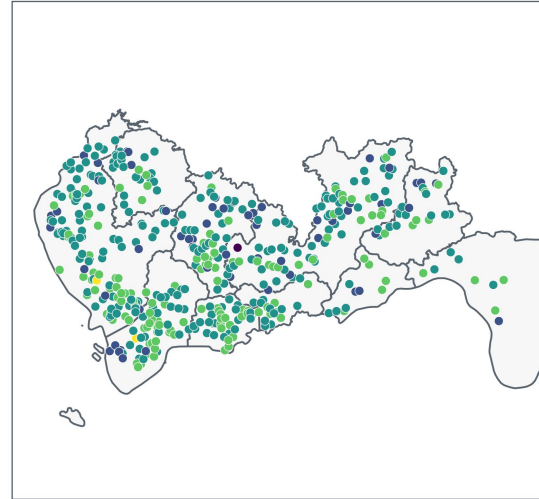
Shenzhen street-view interpretation scores

Two task definitions, one matched 465-location geospatial contract

(a) PRS four-dimension mean



(b) Beautiful



Score (fixed 0-1 scale)

Score classes

- 0.0-0.2
- 0.2-0.4
- 0.4-0.6
- 0.6-0.8
- 0.8-1.0

0 7,500 15,000 m



Figure 7. Restorative-quality four-dimension mean (left) and six-dimension Beautiful score (right) at 465 Shenzhen locations. Both QGIS maps use a fixed zero-to-one scale and EPSG:32649. Global Moran I is 0.098 for the PRS mean and 0.182 for Beautiful; both permutation pseudo-p values are 0.001. No local Low-Low observation remains significant after two-sided testing and Benjamini-Hochberg FDR correction.

The restorative-quality mean shows weak positive spatial autocorrelation under the declared $k = 8$ neighbour matrix, but this association does not establish a causal spatial process. The corrected local result also differs from the preceding analysis: 40 PRS Low-Low candidates appeared under unadjusted local thresholds, while none remain significant under the current FDR rule. The map therefore shows the score distribution and global spatial context without identifying local intervention candidates.

6.5 Framework generality: additional task definitions

Three further task definitions run on the same 465 images and the same live model (gpt-5.4-mini) through the identical pipeline, changing only a YAML file and a prompt. These extensions test whether the restorative-quality data contract, with one row per capture containing coordinates, image identity, physical CV features and semantic VLM fields, supports other semantics and input signatures. They are not parallel case studies with equivalent validation evidence.

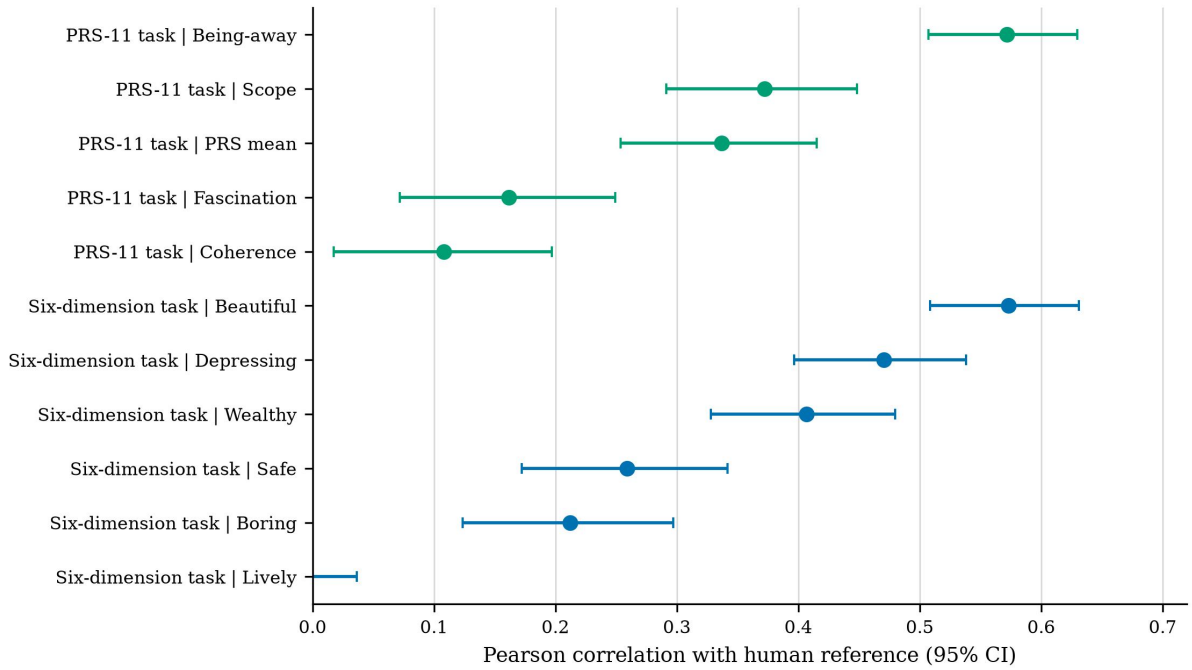


Figure 8. Field-level Pearson correlations for PRS-11 and the six-dimension task. Agreement varies substantially within both tasks and cannot be represented by one accuracy value. Being-away and Beautiful have the highest correlations in their respective tasks; Coherence and Lively have the lowest, and the interval for Lively crosses zero.

urban_perception_sixdim was evaluated against a separate six-field human reference. Five of the six Pearson correlations were significant and positive, ranging from $r = 0.212$ for Boring to $r = 0.573$ for Beautiful; Lively showed no significant relationship ($r = -0.055$, $p = 0.24$). The predictions contained 36 to 66 distinct values per field but retained directional bias, largest for Safe (+0.306) and Depressing (-0.418). Beautiful had a global Moran I of 0.182 (Figure 7), while no corrected local cluster remained significant.

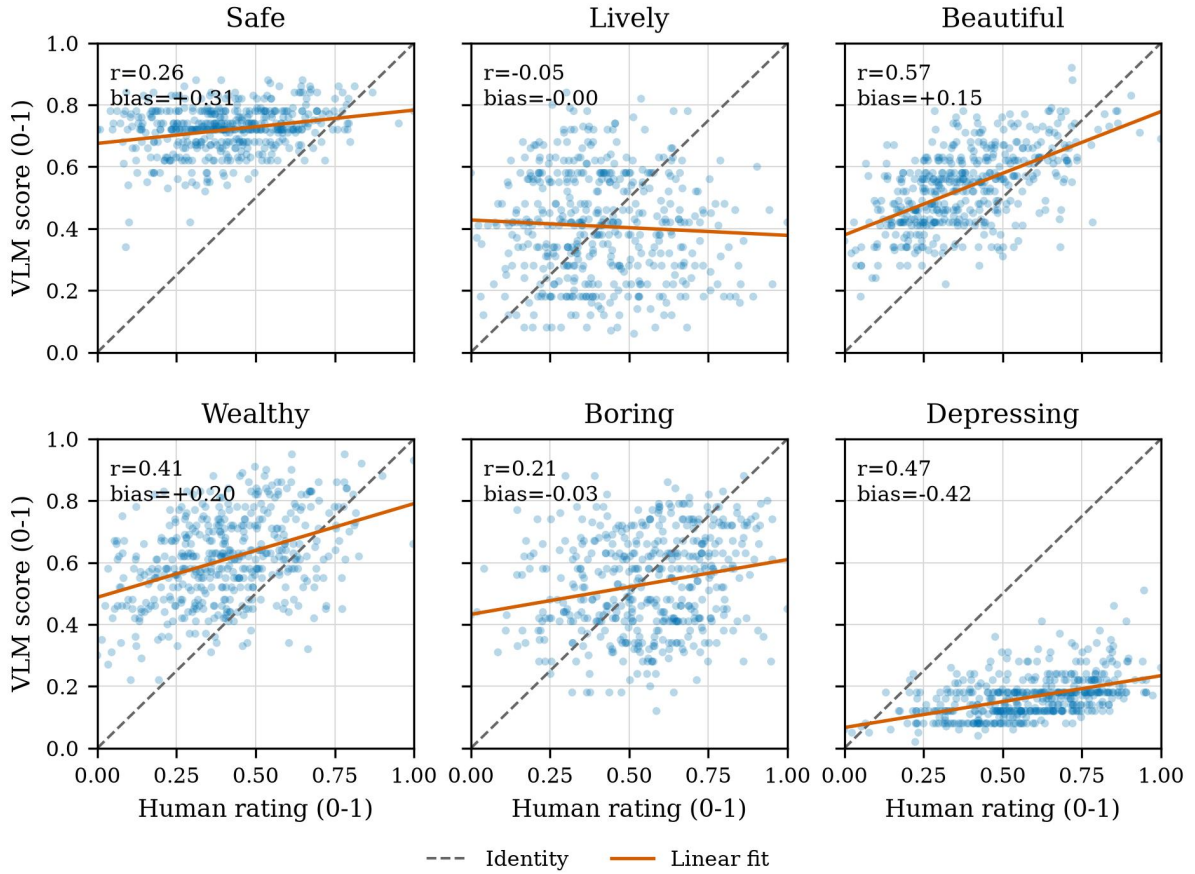


Figure 9. Calibration of six-dimension predictions against human ratings. Safe, Wealthy and Beautiful predictions lie mostly above the identity line, while Depressing lies below it. Correlation and calibration answer different questions: a model can preserve rank ordering while misrepresenting the human scale.

urban_renewal_pair demonstrates that the same pipeline accepts audited before/after image pairs (date/distance/heading auditing, two-leg CV deltas and structured VLM change fields), but only with four synthetic pairs because no same-location, two-date panel exists in the available captures at any audited tolerance. It therefore supports no real detection-accuracy claim.

renewal_priority_screening explores, on the same 465 single-scene captures, whether visible disrepair, neglect, informality or clutter correlates with a low restorative-quality score: it does, moderately ($r = -0.433$, 95% CI [-0.504, -0.356]), consistent with the motivating hypothesis, but no human renewal-priority reference exists, so the task is formally not validated.

The framework already produces a reusable, traceable street-view dataset that a future study with genuine temporal image pairs and human renewal labels could build on. These extensions are not, on their own, a renewal-detection result.

7. Discussion

7.1 What the evaluation establishes

The evaluation supports the project's engineering claim at three levels. Automated tests show that the declared contracts and failure cases behave as specified. The PRS-11 replay shows that historical evidence can enter the same product family without losing source provenance. The six-dimension and renewal task definitions show that a new real semantic task, or a temporal-pair task, can run on the same images without modifying framework code, though only the six-dimension task carries a human reference to validate against.

These levels are deliberately different. A passing test supports software behavior. A schema-valid VLM row supports operational completion. Correlation and calibration support a task-specific comparison with one

human reference. A synthetic pair supports mechanism execution. None of these automatically transfers to another city, model, reference population or policy conclusion.

7.2 Task-dependent validity

The six-dimension results show substantial variation within one prompt and one model. Beautiful reaches $r = 0.573$, while Lively shows no significant relationship at all ($r = -0.055$). The PRS replay shows a comparable spread, from Being-away at 0.572 to Coherence at 0.108. A single statement such as "the VLM agrees with humans" would hide this variation. PyRestore therefore treats validation mapping as part of the task and retains one row per target.

The calibration results add a second limitation. The six-dimension predictions contain 36 to 66 distinct values per field but remain less dispersed than the human references. Safe and Depressing have large biases in opposite directions. A model may therefore preserve some ordering while failing to reproduce the human scale. Raw model values should not be interpreted as direct survey substitutes in planning or comparative monitoring. Local calibration would require an explicit estimation method and held-out evaluation.

A likely contributor is the zero-shot prompt design itself: the task supplies a scoring rubric but no worked examples anchoring the numeric scale to concrete scenes, and none of the retrieval-augmented prior knowledge that Ma and Kwan combined with VLM descriptions for the PRS-11 reference (Section 2.2). Without such an anchor, the model has no external reference for where 0.2 or 0.8 should fall on the human scale, even when it orders scenes correctly. This explanation is consistent with the observed pattern but not established by the present evidence; testing it would require a few-shot or retrieval-augmented variant of the same task run under the same protocol.

The human references also have boundaries. They are existing ratings associated with the Ma and Kwan data package, not a survey newly administered for PyRestore. The matched 465 rows reflect local image availability. The framework exposes these facts through reference type, matching reports and data-status fields, but software provenance cannot remove sampling or cultural limitations.

7.3 Physical and semantic evidence remain separate

CV and VLM outputs describe different evidence families. Segmentation estimates quantify the fraction of visible pixels assigned to classes such as vegetation, sky and building. VLM fields interpret constructs such as Safe, Beautiful or Being-away. A high vegetation share may contribute to a semantic judgement, but the two values are not interchangeable and their validation references differ.

PyRestore coordinates these families without claiming that fusion improves prediction. The paired case attaches GVI, SVF and enclosure deltas as physical-change context next to a semantic renewal decision. It does not combine them into an unvalidated composite. A future coordinated decision rule would require labelled real pairs and an explicit comparison against CV-only and VLM-only baselines.

7.4 Spatial interpretation after error control

The spatial revision illustrates the difference between exploratory display and statistical inference. Global Moran statistics remain positive with small permutation p values. Hundreds of local tests, however, create a multiplicity problem. Applying two-sided local pseudo-p values and BH FDR eliminates every Low-Low candidate at the selected threshold. The supported conclusion is weak global clustering without FDR-supported local clusters.

This result does not show that no low-score area exists. It shows that the present sample, indicator, $k = 8$ weights and multiple-testing rule do not support observation-level significance. Different spatial units, weights or independently designed confirmatory tests could answer different questions. PyRestore stores the weight parameters and adjusted p values so later sensitivity analyses remain distinguishable from the reported result.

7.5 Paired-scene limitations

The renewal task is the strongest demonstration of architectural heterogeneity and the weakest empirical case. The local Shenzhen archive includes many years but does not provide a sufficient same-location panel under the declared tolerances. Synthetic solid-colour pairs cannot express façade replacement, temporary occlusion, viewpoint drift or routine maintenance. They also cannot test the expert constraints derived from Bai et al. [9].

The report therefore limits the claim to software mechanics. A substantive follow-up requires a real panel, a documented sampling frame, at least a validation subset with human pair labels, nuisance negatives, inter- or intra-annotator agreement and provider/model repeatability. Precision, recall, F1, abstention coverage and false-positive rate on nuisance negatives would then become valid outputs.

7.6 Software trade-offs

The framework favours inspectability over maximum throughput. VLM requests run sequentially, with caching and resumable batch scripts controlling cost and failure recovery. SQLite is appropriate for one-machine research runs but is not a distributed execution store. The task schema supports the field types consumed by the four implemented tasks rather than a complete JSON Schema engine. Only single and paired SceneSets are implemented; longer sequences remain future work.

Provider aliases and availability can change. The failed requested deployment and successful fallback demonstrate why compatibility testing belongs in the run protocol. Although request fingerprints identify declared settings, they cannot guarantee that an external provider's mutable model alias always resolves to identical weights. Cached responses preserve replayability after the first call; they do not make the original model call deterministic.

The optional CV stack is also substantial. PyTorch, Transformers and OpenCV are separated from core dependencies, which keeps offline tests and precomputed workflows lightweight. This design means default CI does not execute the live segmentation model. A release process intended for wider deployment should add a small licensed image fixture and a scheduled optional-dependency smoke test.

8. Conclusions

This project develops PyRestore, a configurable framework that transforms standardized street-view inputs into analysis-ready datasets. Researchers provide a georeferenced image manifest and a task configuration that states the perception question to be analyzed. The framework extracts CV-derived physical scene indicators and VLM-derived perception indicators, then organizes them into spatial data that can be used for GIS mapping, statistical tests and later modelling. Single images and before-and-after image pairs use the same processing workflow, so each study does not need to maintain its own pipeline.

The Shenzhen restorative-quality case illustrates this workflow. The historical PRS-11 scores reached $r = 0.337$ against human ratings for the four-dimension mean. Spatial analysis found weak global autocorrelation but no FDR-significant local Low-Low candidates. A six-dimension task further showed that agreement and calibration can vary substantially across perception indicators. The paired renewal task validates processing mechanics on synthetic data only, while the single-scene renewal-priority task remains exploratory because no human reference is available.

PyRestore does not introduce a new perception model. Its contribution is to extend street-view research beyond physical scene measurement by producing datasets that also contain visual-perception indicators. Researchers can use task configurations to introduce different questions, prompts, examples and references for studies of restorativeness, safety, beauty or other human-centred urban outcomes. Renewal research still requires comparable temporal imagery and human labels, and each new task requires independent validation.

The project also relates to two of the Sustainable Development Goals named for this course. Its most direct relevance is to Goal 11: it provides a reproducible way to organise street-level evidence for public-space conditions, and the absence of FDR-significant local candidates shows why an exploratory map should not be converted directly into an intervention list [16]. It relates to Goal 9 through reusable, interoperable research-

software infrastructure (a locked environment, explicit task and data contracts, machine-readable provenance and GeoPackage products) aligned with practical FAIR4RS and OGC principles [14, 15].

9. User Manual

PyRestore uses Python functions and Jupyter notebooks as its public interface. An analyst prepares a manifest with stable identities, paths and coordinates, selects a task YAML and calls `run_task`. Precomputed analyzer tables may be supplied for an offline replay; live VLM execution additionally requires an explicit remote-image authorization setting. The analyst inspects `quality_report.csv` and analyzer error fields before using `indicators.gpkg` in QGIS. Validation results apply only to the declared reference type and matched records.

The complete operational manual is published in the project's GitHub repository ([<https://github.com/GuojialeGeographer/2026GProject.git>]/`USER_MANUAL.md`) alongside the source code.

10. Repository structure

- `pyrestore/`: reusable Python package;
- `tasks/`: four declarative task definitions and prompt resources;
- `notebooks/01-04`: single-scene, Shenzhen PRS, add-a-task and paired-scene demonstrations;
- `outputs/submission/`: frozen real and synthetic evidence products;
- `submission/assets/`: editable Draw.io diagrams, vector result figures and QGIS project/map exports;
- `docs/`: requirements, architecture, terminology, protocols, evidence ledger, report and user manual;
- `tests/`: 75 offline automated tests;
- `uv.lock`: reproducible dependency resolution;
- this report and the presentation, submitted through WeBeep; the user manual is published in the GitHub repository rather than delivered as a separate submission file.

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